EcoPulse

A SIMPLE GUIDE

How EcoPulse works

From one photo on your phone to a problem fixed in your street. What you do, what happens next, and how the city team acts on it.

1. **You report** — Photo + location in about a minute
2. **App checks** — Joins duplicates, judges how urgent
3. **Neighbours confirm** — "Yes, I see it too"
4. **City team fixes** — Most urgent first
5. **You're told** — And can check the fix is real

Inside: what EcoPulse is · reporting a problem step by step · what happens after you press Send · how the city team works · your privacy · how to start the app on a computer · words explained

THE IDEA

What is EcoPulse?

EcoPulse is a free app that lets anyone tell the city about pollution and civic problems near them, like garbage, sewage, smoke, noise or flooding, and then follow what is being done about it.

You don't need any special equipment. A phone with a camera and GPS is enough. Reports from many people are put together so the city can see where the problems are and fix the worst ones first.

Who uses it

Residents

Report problems, confirm problems others reported, see how their area is doing. Uses the **Citizen app** on a phone.

City staff

See a ranked list of problems, send teams, mark work started and fixed. Uses the **Command Center** on a computer.

Everyone

Sees a live map, area scores and a feed of what's been reported and fixed across the city.

What you can report

Problem	Example	Problem	Example
Garbage	Overflowing bins, litter	Building dust	Dust from construction
Dumped waste	Rubble or trash on land	Noise	Loudspeakers, horns, machines
Sewage	Overflowing or open sewage	Smoke or bad air	Fumes, bad smell, smog
Blocked drain	Clogged or broken drains	Flooding	Flooded roads or homes
Standing water	Water collected on the road	Dying trees	Dead or fallen trees
Burning waste	Smoke from burning trash	Tree cutting	Trees being cut down
Plastic waste	Piles of plastic, bags	Dead animal	Dead animal on the street
Dirty water	Polluted lake, canal or tap	Old electronics	Dumped phones, batteries

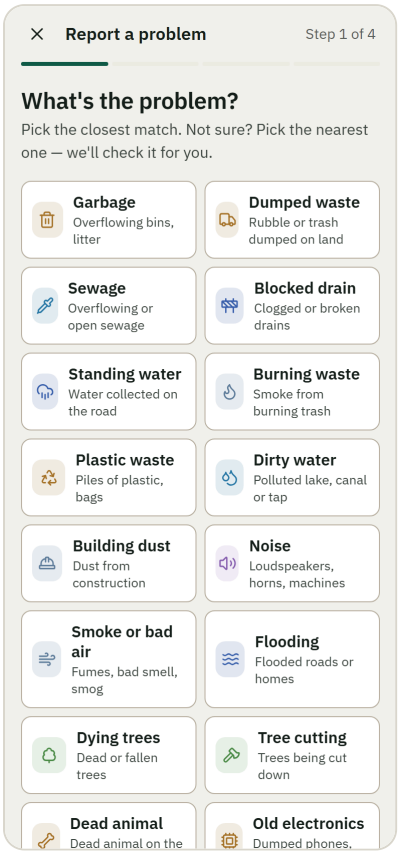
Not sure which one? Pick the closest. The app looks at your photo and suggests a better match if it thinks you picked the wrong one.

No sign-up needed. Tap *Continue as guest* to start. You can make an account later so you keep your reports and points if you change phones.

PART 1 · WHAT YOU DO

Reporting a problem, step by step

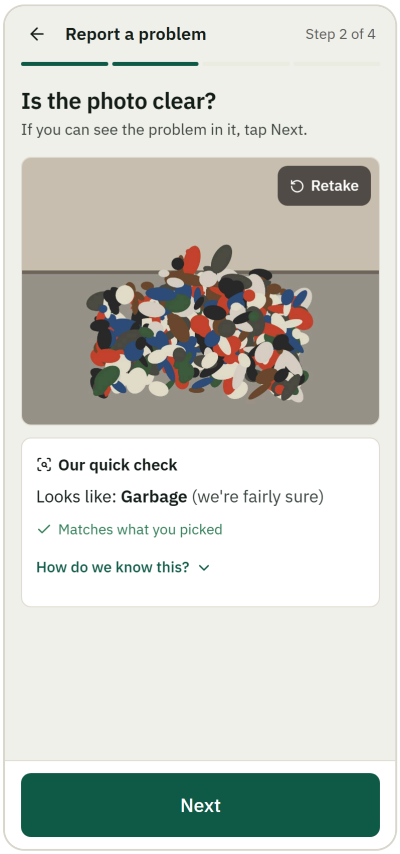
Tap the big green **Report** button at the bottom of any screen. There are four short steps.



Step 1 of 4

What's the problem?

Tap the picture that matches best. Each button has a short example underneath so you know what it means.

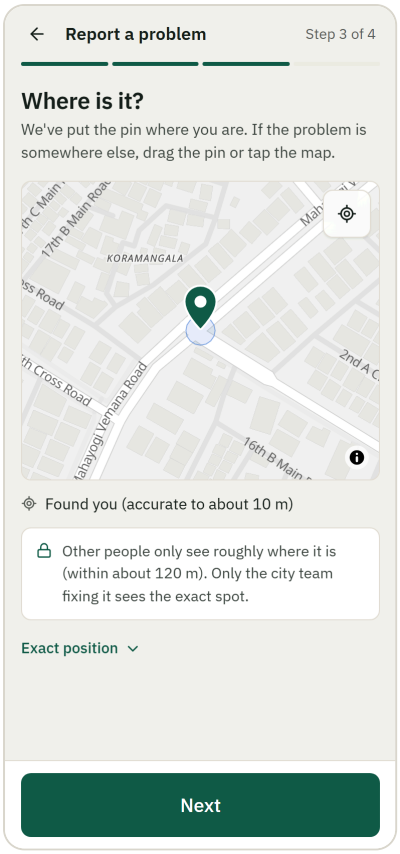


Step 2 of 4

Take a photo

Take a photo or pick one from your gallery. The app looks at it straight away and tells you what it thinks the problem is.

A photo isn't needed for things you can't photograph, like noise or bad smells. For noise, the app can measure loudness using your phone's microphone. Only the number is sent, never a recording.

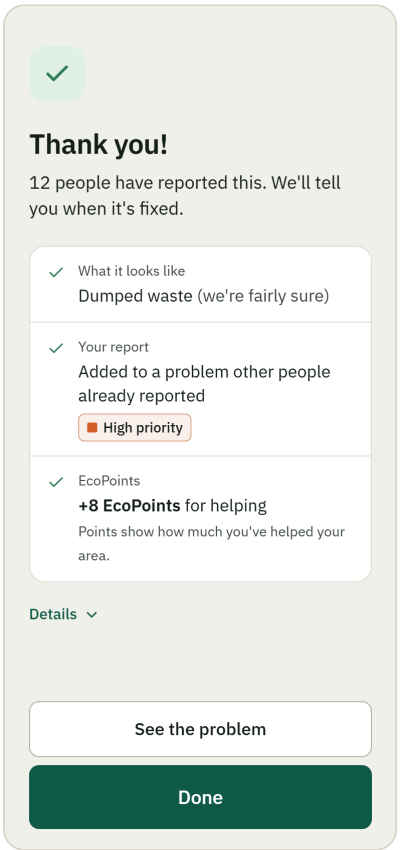


Step 3 of 4

Where is it?

The pin is placed where you are standing. If the problem is somewhere else, drag the pin or tap the map.

Other people only see roughly where it is (within about 120 m). Only the city team fixing it sees the exact spot.



Step 4 of 4 · then Send

Add details (optional) and send

Add a short note if it helps, and choose whether to hide your name. Press **Send**.

You'll see **Thank you!** with what the app thinks the problem is, whether other people already reported it, and the **EcoPoints** you earned.

No internet? Your report is saved on the phone and sent automatically as soon as you're back online.

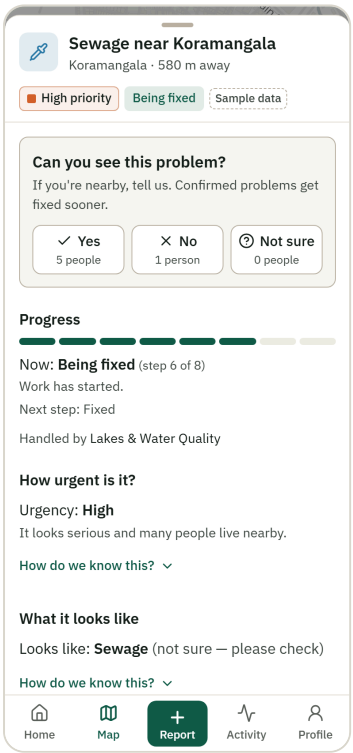
PART 2 · WHAT HAPPENS NEXT

After you press Send

Within a second or two the app does the following on its own:

1. Check the type	The photo is checked to confirm the kind of problem. If it isn't sure, it says so ("not sure, please check") instead of guessing.
2. Join duplicates	If someone already reported the same kind of problem very close by, your report is **added to theirs** instead of creating a new one. "12 people reported this" means 12 separate reports were joined.
3. Judge urgency	Each problem gets an urgency: **Urgent, High, Medium** or **Low**. It depends on how serious it looks, how many people reported it, how long it has been open, how many people live nearby, whether a school or hospital is close, and today's air or weather readings.
4. Tell everyone nearby	The problem appears on the map, in the Activity feed and on the city team's screen straight away, without anyone refreshing.
5. Update area scores	The neighbourhood score and the "problem areas" (places where many reports cluster) are recalculated.

Neighbours confirm it



When people nearby open a problem, they are asked: **Can you see this problem?** They can answer **Yes**, **No** or **Not sure**.

When **3 or more** people say "Yes" (and clearly more say Yes than No), it is marked **Confirmed** and moves up the city's to-do list. "No" answers lower the app's confidence that the problem is real.

The same screen shows the **Progress** bar, how urgent the problem is and why, and photos from before and after the fix.

PART 3 · GETTING IT FIXED

The life of a problem

Every problem moves through these steps. You can see which step it's on at any time.

Received → Checked → Confirmed → On the to-do list → Team assigned → Being fixed → Fixed → Fix confirmed by neighbours

Step	What it means
Received	We have the report.
Checked	The app has checked what kind of problem it is.
Confirmed	Neighbours or staff confirmed it is real.
On the to-do list	It's on the city's list of things to fix, ranked by urgency.
Team assigned	A city team (for example Waste, or Lakes & Water) is responsible for it.
Being fixed	Work has started.
Fixed	The city says it's fixed and adds an "after" photo.
Fix confirmed	3 or more people nearby confirmed it's really gone. If 3 or more say it's still there, it is opened again.

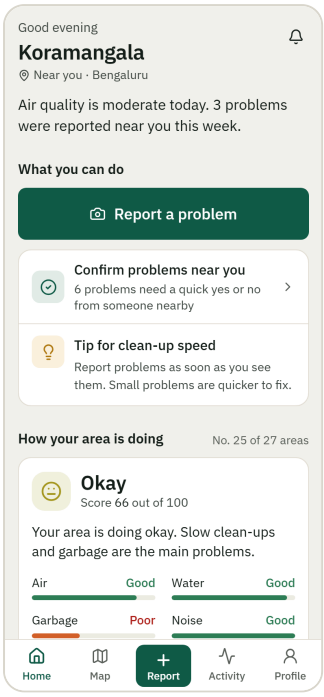
What the city team does (Command Center)

1. Staff see a **priority list**, with the most urgent problems at the top, next to a live map of the city.
2. They open a problem to see the photos, how many people reported it, the timeline, and **why** it was ranked that way.
3. They **confirm** it, **assign** a team, mark **work started**, and mark it **fixed** with an after-photo.
4. They also see **problem areas** (places where reports cluster, including ones expected to get worse), **suggestions** such as a new waste point or tree line, and charts showing how fast problems are being fixed.

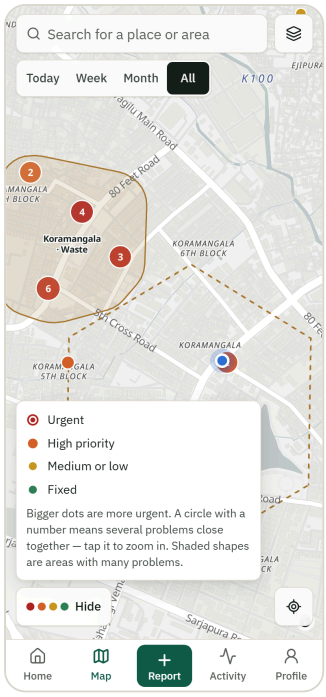
You're kept in the loop. Every time your problem moves forward, for example when it's confirmed, assigned or fixed, you get a message in the app (the bell icon on Home).

PART 4 · USING THE APP DAY TO DAY

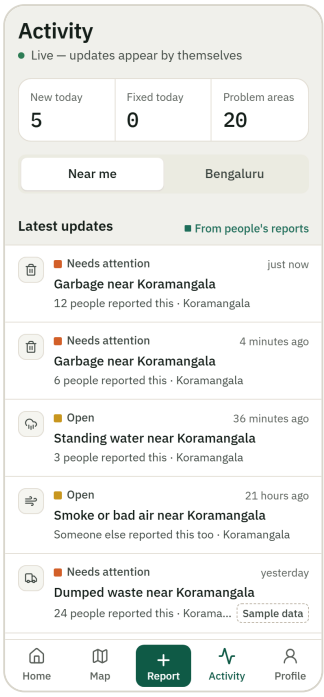
The five tabs at the bottom



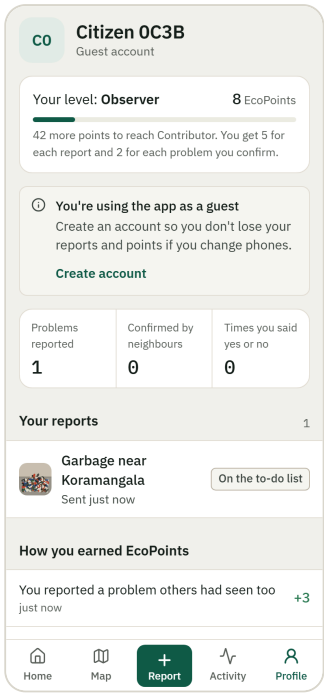
Home: how your area is doing, problems near you



Map: every problem as a coloured dot



Activity: live feed of new and fixed problems



Profile: your reports, points and settings

Tab	What it's for
Home	A one-line summary ("Air quality is moderate today. 3 problems were reported near you this week."), your area's score out of 100 with Good, Okay or Poor for air, water, garbage, noise, trees and clean-up speed, and a tip for your area.
Map	Red means urgent, orange high, yellow medium or low, green fixed. A circle with a number is a group of problems close together; tap it to zoom in. Use *Today / Week / Month* to filter.
Report	The big green button. It opens the 4-step report.
Activity	What's happening near you or across the whole city, updated live.
Profile	Your reports and their status, your EcoPoints and level, text size, dark mode, notifications and help.

EcoPoints: 5 points for each report and 2 for each problem you confirm. Points show how much you've helped your area; levels go from Observer upwards.

Easier to read: Profile → Settings lets you make the text **larger** or switch to **dark mode**. **Profile → Help** answers common questions.

PART 5 · TRUST

Your privacy and honest data

Your privacy

- **Your exact spot is never public.** Other people only see roughly where a report is (within about 120 m). Only the city team fixing it sees the exact point.
- **Photos are cleaned.** Hidden location and camera details are removed from every photo before it is stored.
- **Noise stays on your phone.** Only the loudness number is sent, never the sound.
- **You can report anonymously.** Turn on "hide my name" and even city staff won't see who sent it.
- **Location only while the app is open.** It is never tracked in the background.

Honest about where numbers come from

The app always tells you where information comes from:

Label	Means
From people's reports	What residents actually reported.
From public data	Open data such as today's air quality and weather (Open-Meteo) and maps of parks, schools and hospitals (OpenStreetMap).
Our estimate	Calculated by the app, such as area scores and urgency. Tap "How do we know this?" to see how.
A suggestion, not a fact	Ideas for the city, such as where a new waste point could help.
Sample data	Example data used for the demo. It is marked so it is never mistaken for real reports.

Good to know: scores come from what people report, so an area with very few reports can look better than it really is. The more people use it, the more accurate it gets.

PART 6 · RUNNING IT YOURSELF

How to start EcoPulse on a Windows computer

No coding needed. You only do the first two steps once.

1. Install **Python 3.12** from python.org (tick "*Add to PATH*" during install) and **Node.js LTS** from nodejs.org.
2. In the EcoPulse folder, double-click `setup.bat`. This takes about 5 to 10 minutes, needs internet, and loads a demo city.
3. Double-click `start.bat`. Two pages open in your browser:
   - **Citizen app:** `http://localhost:5174`. Tap *Continue as guest*.
   - **Command Center** (city staff): `http://localhost:5173`. Sign in with `officer@ecopulse.dev` and password `ecopulse-demo`.
4. When you're finished, double-click `stop.bat`.

If something goes wrong, the scripts tell you what to do, and details are saved in the `.logs` folder. The citizen app is designed for a phone; on a computer it shows in a phone-sized column.

Words explained

Word	Meaning
Problem	One issue in one place. Several people's reports of the same thing are joined into one problem.
Confirm	Saying "yes, I can see it too", or later "yes, it's really fixed".
Urgency	Urgent, High, Medium or Low: how quickly the city should act.
Problem area	A place where many problems are reported close together.
Area score	0 to 100 for your neighbourhood; higher is better. Good is 75 or more, Okay is 55 to 74, and Poor is below 55.
Air quality (AQI)	A number for how clean the air is today: good, moderate, unhealthy for sensitive people, unhealthy or very unhealthy.
EcoPoints	Points you earn for reporting and confirming problems.
Guest	Using the app without an account. If you create an account later, you keep your reports and points.